



ST BENEDICT'S COLLEGE

SUBJECT	LIFE SCIENCES	DATE	24 NOVEMBER 2022
GRADE	11	MARKS	170
EXAMINER	MRS S. STEGMANN	MODERATOR	MRS B. ZAJAC
		DURATION	3 HOURS

MARKING GUIDELINES

LIFE SCIENCES

Question 1:

1.1.

1.1.1	1.1.2	1.1.3	1.1.4	1.1.5	1.1.6	1.1.7	1.1.8	1.1.9	1.1.10
B ✓	C ✓	B ✓	B ✓	A ✓	C ✓	C ✓	B ✓	A ✓	C ✓

[10]

1.2. F ✓

A ✓

C ✓

J ✓

K ✓

H ✓

E ✓

I ✓

D ✓

B ✓

[10]

1.3.

1.3.1. A- Bacteria ✓ C- Fungi ✓

(2)

1.3.2. 1- Flagellum ✓ 2- Nucleic acid (DNA/ RNA) ✓ 3-Sporangium ✓

(3)

1.3.3.

a. B ✓

(1)

b. C ✓

(1)

1.3.4. Bacillus (rod shaped) ✓

(1)

1.3.5. Heading ✓

Accuracy ✓

Labels x3 ✓✓✓

(5)

[13]

LIFE SCIENCES

- 1.4.
- 1.4.1. Growth of bacteria / diameter of the area with no bacterial growth ✓ (1)
- 1.4.2. Same strain of bacteria
- Same amount of bacteria in all the petri dishes
- Same amount of agar medium
- ALL measurements to be done by the same person with the same instrument ✓✓✓ (any 3)
- 1.4.3. For increased reliability ✓ (1)
- 1.4.4. Acts as a control- to verify that any difference in bacterial growth is due to the action of Vibramycin ✓ (1)
- 1.4.5.
- a. An increase in the concentration of Vibramycin antibiotic will increase bacterial growth ✓✓
- OR**
- An increase in the concentration of Vibramycin antibiotic will decrease bacterial growth ✓✓
- OR**
- An increase in the concentration of Vibramycin antibiotic will have no effect on bacterial growth ✓✓ (2)
- b. Concentration of Vibramycin ✓ (1)
- [9]**
- 1.5.
- 1.5.1. Reflex arc ✓ (1)
- 1.5.2. To bring about a quick response ✓ to minimize danger/ protection mechanism ✓ (2)
- 1.5.3.
- a. B: motor / multipolar / efferent ✓
- b. C: interneuron / connector neuron ✓
- c. E: sensory / unipolar / afferent ✓ (3)
- 1.5.4. a. F ✓ b. A ✓ (2)
- 1.5.5. a. D ✓ - synapse ✓ b. G ✓ - myelin sheath ✓ (4)
- 1.5.6. The heat/pain would be felt ✓ but no response would be brought about / response would occur slowly by alternate mechanisms ✓ (2)

[14]

LIFE SCIENCES

1.6.

1.6.1. Mark-recapture method ✓ (1)

1.6.2. Indirect ✓ (1)

1.6.3. An estimation of the population size is given✓, not the exact number ✓ (2)

1.6.4. No ✓

Slow moving therefore quadrat method could be done.

In a garden, which is a small area, therefore quadrat method would be more accurate. ✓

OR

Yes ✓ Valid reason ✓ (2)

1.6.5.

$$P = \frac{M \times C}{R}$$

R

$$M = \frac{P \times R}{C}$$

C

$$= \frac{60 \times 5}{15} \checkmark$$

$$= 20 \checkmark$$

$$= 20 \text{ snails } \checkmark \text{ in first catch} \quad (3)$$

1.6.6. Do not harm the organisms; Do not hinder the movement of organisms; Allow organisms to mingle freely with the rest of the population; Do not let too much time pass between captures
✓✓ (any 2)

1.7. C ✓ A ✓ B ✓ [3]

TOTAL QUESTION 1: 70 MARKS

LIFE SCIENCES

Question 2:

- 2.1.1. A-protein coat ✓ B- RNA/ nucleic acid ✓ (2)
- 2.1.2. Antibiotics are used to kill/ fight off bacteria (living organisms). Viruses are not living. ✓ (2)
- 2.1.3. A person is given a weak strain of the germ. ✓
 The body will produce antibodies to fight the infection ✓
 The antibodies will protect them against a new/ stronger infection of the same germ. ✓ (3)
- 2.1.4. It would need to go through a trial✓ to ensure it has no negative effects ✓ (2)
- 2.1.5.
- 2.1.5.1. 1960- 1930 = 30 years ✓✓ (28-32 accepted) (2)
- 2.1.5.2. Polio ✓ Swine flu ✓ (2)
- 2.1.5.3. The time from first studies to roll out and eradication decreases over time. ✓✓ (2)
- 2.1.5.4. There is an advance in medicine
 More information is known about microorganisms
 Technology to study microorganisms has improved greatly
 More scientists are involved in the process of producing vaccines
 Any reasonable answer ✓✓ (any 2)
- 2.1.5.5. Cooler conditions are more favourable for viral spread
 Lots of people remain indoors and closer to others making the spread easier
 Any other answer ✓✓ (any 2)

[19]

- 2.2.
- 2.2.1. To determine the effects of antibiotics on the treatment/ growth of TB ✓✓ (2)
- 2.2.2. Bacteria ✓ (1)
- 2.2.3. Antibiotic 3 fights of the TB bacterium ✓
 Antibiotic 2 lowers the growth rate of the TB bacterium ✓
 Antibiotic 1 has no effect on the growth rate of the TB bacterium/ TB bacterium is resistant to antibiotic 1. ✓ (3)
- 2.2.4. With a long (6 month) ✓ course of antibiotics ✓ (2)
- 2.2.5. Overcrowded living spaces more prominent
 Poorly ventilated living spaces

LIFE SCIENCES

Poor diet

Poor health/ more diseases such as HIV prevalent therefore less immunity

More difficult to access medical treatment

Lack of education/ awareness ✓✓✓

(any 3)

[11]

Total Question 2: 30 marks

Question 3:

3.1. The number of lady birds and green flies in an ecosystem were counted periodically from J

3.1.1. Heading ✓

Key ✓

Plotting ✓✓

Joining ✓

(5)

3.1.2. Every 3 months ✓

(1)

3.1.3. Predator-prey / predation ✓

(1)

3.1.4. They are the predator.

(1)

3.1.5. Their numbers are consistently lower than that of the green fly population ✓

Their numbers rise and fall after the rise and fall of the green flies ✓

(2)

3.1.6. Sharks typically hunt and kill the weak and injured of the seal population ✓

This allows for stronger seal individual to survive and reproduce ✓

Resulting in a stronger population developing ✓

The stronger and fitter sharks are those that hunt successfully ✓

As a result the strongest sharks survive to reproduce making a stronger shark population ✓

In both populations the weaker individuals, and weaker genes, are eradicated first ✓ (any 4)

3.1.7. The maximum number of individuals/population density ✓an ecosystem can sustain/support ✓

(2)

[16]

3.2.

3.2.1. Interspecific competition ✓

(1)

3.2.2. Beak shapes and sizes ✓

(1)

3.2.3.

a. C ✓

b. A ✓

(2)

LIFE SCIENCES

3.2.4. Yes. ✓

They occupy different ecological niches

The large tree finch eats mainly animal material, while the large ground finch eats mainly plant material. (any 1 ✓) (2)

3.2.5. The large tree finch. ✓✓

Both eat mainly animal material ✓ (3)

[9]

3.3. Growth was stable/ logistical until 1850. ✓

It then increased exponentially until 2017 ✓

This could be due to:

- Advances in the medical field for treating diseases
- Development of vaccinations against infectious diseases
- Increase in cultivation of disease-resistant crops
- Use of artificial fertilizers to improve agriculture
- More effective farming methods to increase food security
- Increase in education regarding medicine, technology and agriculture
- Any other reasonable answer (any 3 ✓✓✓) [5]

Total Question 3: 40 Marks

LIFE SCIENCES

Question 4:

SOURCE	YES	NO
A		Activin-A switches on the remyelination process Binds to a specific receptor and causes cells to carry out myelin repair. Remyelination crucial to slowing down the progress of the disease
B		Disease-modifying drugs' side effects are linked to lowered immune system, not nervous system function. WBC is lowered, but neurons are not directly affected.
C		Ocrelizumab approved to slow the progression of MS. In the process of being made available to the NHS for distribution.
D	Drugs may slow down or speed up the CNS and autonomic functions such as blood pressure, respiration, HR and body temperature. Therefore the reflex center is affected by these drugs. Neurotransmitter levels are also impacted: - Dopamine - Serotonin - GABA - Norepinephrine Brain stem, limbic system and cerebral cortex all affected. This affects life -sustaining functions (HR, breathing, etc) Limbic control over emotions and happiness are affected. "thinking centre" affected therefore planning, decision-making and information processing are affected. NOTE: own info linked to what the change in neurotransmitter levels can do to the conduction of impulses. Can also mention the drugs used are mind-altering and can lead to addiction due to the changing of the brain chemistry. This will affect nervous conduction.	
E	Drugs of abuse interfere with neurotransmission in the 'reward' region of the brain.	

	They alter neurotransmission by neurons that release dopamine. Resulting in communication altering between neurons. Synaptic transmission is affected. NOTE: own info mentioning the function of dopamine and how altering it's release will affect nervous conduction. If receptors are activated or blocked the conduction of impulses will be affected. Most drugs, even mild ones have an effect on neurotransmitter release/ inhibition and reuptake.	
F	Shows reflex path- can be used in description	Shows reflex path- can be used in description
G	Stem cells indicate positive changes in MS patients' immune systems. This therapy along with improved gut health should be used as opposed to mind-altering drugs/medication. Therapies involving gut microbiome, stem cells, neuroprotective treatments, remyelination and MYMD are in the earliest stages and should be first choice in MS treatment. **can argue that further damage to myelin sheath can be stopped, but lowered immune systems could result in other severe illnesses affecting patients.	Cladribine and Siponimid target lymphocytes making a person more at risk of infections. Reduces responses to vaccinations. - Still does not affect nervous conduction directly There is hope that adequate responses to Covid-19 vaccines may still occur. Stem cells indicate positive changes in MS patients' immune systems.

SK- 1-2 (2marks) 3-4 (4 marks) 5-6 (6 marks) 7up (8 marks)

Total Question 4: 40 Marks

Total: 170 Marks